



Presented to:
GlaxoSmithKline

Start-up Document
A Line of Therapy Algorithm for Multiple Myeloma, Using Healthcare Claims, Update

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1 Background and Rationale

Multiple myeloma (MM) is characterized by a proliferation of malignant plasma cells in the bone marrow which causes production of monoclonal immunoglobulins, leading to hypercalcemia, anemia, renal disease, impaired immune function, and bone lesions.^{1,2} MM is a relatively rare disease contributing to 1.8% of all new cancer cases with an estimated 34,470 incident cases and 12,640 deaths in the United States (US) in 2022.³

Treatment of MM is initiated when the patients are symptomatic, or have active disease based on the International Myeloma Working Group (IMWG) definition.⁴ Standard frontline therapy is decided by Revised International Staging System (R-ISS) risk stratification based on multiple criteria including serum beta-2 microglobulin, serum albumin, LDH (lactate dehydrogenase), chromosomal abnormalities by FISH (fluorescence in-situ hybridization, extra medullary disease, frailty, high plasma cell proliferation, renal failure, high serum FLC (free light chains), thrombocytopenia, leukopenia, and eligibility for hematopoietic cell transplantation (HSCT).^{4,5} For transplant-ineligible patients, the preferred first-line treatment regimens are lenalidomide-bortezomib-dexamethasone or daratumumab-lenalidomide-dexamethasone-based regimens; other regimens are recommended or useful in certain circumstances.^{5,6} With an enhanced understanding of the pathophysiology and disease heterogeneity, and the advent of novel treatments, there have been improvements in overall survival of MM patients.⁶

Despite the positive trends in disease survival, the natural evolution of MM results in relapse after initial therapy.⁷ The choice of treatment from various available regimens to treat relapsed and refractory MM (RRMM) is based on numerous factors. There are no firm treatment guidelines as clinical decisions are based on toxicity, patient and physician preferences, prior treatment, cytogenetic risk, comorbidities, myeloma markers and financial considerations.^{8,9,10} Treatment options for RRMM patients include re-administration of initial agents for those who are naïve or sensitive to the novel drug combinations and were not exposed to them in the previous six months. Patients with disease refractory to the novel drugs should be considered for regimens that do not contain the drugs that they progressed on. Other considerations for RRMM include newer agents in the same class (carfilzomib, pomalidomide), novel mechanisms of action drugs (XPO inhibitors, BCL-2 inhibitors, E3 ligase modulators, new alkylators, immunotherapy based

on conjugated anti-BCMA antibodies and CAR-T cells), combination therapies and autologous/allogeneic HCTs.¹¹ The complexity of the MM treatment landscape, due to multiple treatment phases, gaps between regimens, and combination therapies of three or more drugs, has made it more difficult to identify the real-world clinical pathways of the patients.^{11,12,13}

Identifying the prior lines of therapies (LOT) can support selection of treatment regimens for relapsed MM patients, determine eligibility for enrolment into clinical trials, support adherence, and enable accurate health technology assessment for approvals (e.g., FDA).^{14,15} As additional novel therapies for MM become available, particularly in the second- and third lines, there is a need to understand various lines of therapy for RRMM patients. The goal of this project is to develop a set of rules to identify lines of therapy for MM patients using administrative claims data and validate the rules by linking the claims to electronic health record (EHR) data.

2 MM LOT Algorithm Version

This document describes the second (and, for now, the presumptive final) iteration of the MM LOT algorithm that GSK and Optum have developed. Optum implemented previous versions of the LOT algorithm, which is documented in the first version of the start-up document and data specifications. The output of the first iteration of the MM LOT algorithm was analyzed with EHR data and presented to GSK's external experts (EES). EES feedback led to revisions in the first version of the LOT algorithm, principally, but not limited to, exclusion of corticosteroids by themselves as an MM treatment.

3 Study Purpose and Objectives

The purpose of this project is to develop and validate a LOT algorithm for MM to be used for administrative claims data. The two project objectives are to:

- 1) Develop a LOT algorithm to identify up to five LOTs for MM using administrative claims data.
- 2) Validate the LOT algorithm developed using treatment data from the EHR.

4 Study Design and Plan

4.1. Study Design

This retrospective study of newly diagnosed multiple myeloma health patients uses Optum's Market Clarity integrated administrative claims and EHR data for the period of 1 July 2015 to 30 September 2022 (study period). These are the most recent available Market Clarity data at the time of study initiation. Patients with newly diagnosed MM are identified from 1 January 2016 to 31 March 2022 (identification period). The date of the first claim with a diagnosis code for MM during the identification period is the index date. The baseline is the period of 6 months before the index date. The variable-length follow-up begins on the index date and ends on the 15th day of the month and year of death for patients with month/year of death < 6 months after the index date; otherwise, for patients who do not have an assigned date of death < 6 months after the index date, the variable follow-up is ≥ 6 months long and ends at the earliest of death, health plan disenrollment or the end of the study period.

Starting the study period in 2015 and ending in 2022 will provide more than five years of data for patients with index dates earlier in the identification period, which is a sufficiently long study period to capture patients who are newly diagnosed and develop MM in the relapse/refractory setting.

4.2. Data Sources

4.2.1. Market Clarity Dataset

Data for this study come from the Optum® Market Clarity Integrated Clinical + Claims Database (Market Clarity). Market Clarity provides claims and EHR data that are fully HIPAA-compliant, statistician-certified, and de-identified. Market Clarity links historical EHR data with administrative claims data, i.e., pharmacy claims, physician claims, and facility claims, as well as health plan enrollment information. The linkage between the claims and EHR is deterministic and is based on combinations of patient name, social security number, and date of birth. Data elements sourced from the EHR include lab results, vital signs and measurements, diagnoses, procedures, written prescriptions, and information derived from unstructured clinical notes using natural language processing (NLP).

The Market Clarity database from which the current study sample is identified comprises more than 76 million patients with both EHR and medical/pharmacy claims information. Demographics of patients in Market Clarity are presented below in Table 1.

Table 1. Characteristics of Patients in Market Clarity and Population in US Census

Characteristic	Market Clarity ⁱⁿ (Jan 2007 – Jun 2022)		US Census Population ⁱⁱ (2021)	
	Number of patients (million)	%	Number of people (million)	%
Total	76.8	100%	328.1	100%
Insurance Type				
Commercial	48.8	64%	216.4	66%
Medicaid	13.5	18%	61.9	19%
Medicare	13.4	17%	60.2	18%
Medicare Advantage + Part D	n/a	-	n/a	-
Other Payor Type	4.8	6%	3.2	1%
Uninsured	15.0	20%	27.2	8%
Unknown	10.1	13%	n/a	-
Census Region				
Midwest	30.4	40%	67.9	21%
Northeast	13.9	18%	56.1	17%
South	17.2	22%	126.1	38%
West	11.1	14%	77.9	24%
Unknown ^{iv}	4.1	5%	n/a	n/a
Age Cohort (years) ⁱⁱⁱ				
0-18	10.2	13%	77.0	23%
19-25	5.6	7%	29.0	9%
26-34	9.3	12%	40.3	12%
35-54	20.2	26%	125.5	38%
55-64	10.6	14%	56.2	17%
65+	20.9	27%	56.2	17%
Unknown ^{iv}	0.0	0%	n/a	n/a
Gender				
Female	41.0	53%	166.2	51%
Male	35.8	47%	160.0	49%
Unknown	0.0	0%	n/a	n/a
Race/Ethnicity				
African American	7.4	10%	44.0	13%
Asian	1.8	2%	20.7	6%
Caucasian	45.8	60%	194.2	59%
Hispanic	4.5	6%	62.5	19%
Other/Unknown	17.3	22%	6.7	2%

ⁱⁿ The Optum[®] de-identified Market Clarity dataset
ⁱⁱ Census: Health Insurance Coverage in the United States: 2022 Health Insurance Coverage in the United States: 2021 (census.gov)
ⁱⁱⁱ Age is calculated as 2022 minus year of birth.
^{iv} Unknown reflects instances when data fell outside of the defined range.

4.2.2. Mortality Data

Data on mortality in the Optum Research Database and Market Clarity are captured from various sources to provide the most complete information on death available. As noted

in Sections 4.1 and 4.3.1.1, assigned date of death was used to as a criterion to end the variable follow-up period.

Mortality data sources include:

- Social Security Administration (SSA) Death Master File (DMF).
 - Mortality information sourced from SSA DMF are incomplete because some states discontinued disclosing mortality data to the SSA on and after 2011. However, the other mortality data sources below, particularly for Medicare beneficiaries, substantially fill that gap.
- Patient discharge status from facility claims.
- Diagnosis and procedure codes.
- Enrollment records with reason for disenrollment = death; this information comes from enrollees' family members or employers who notify the health plans to disenroll the deceased.
- The Center for Medicare and Medicaid Services (CMS) for patients enrolled in Medicare Advantage plans; these mortality data are relatively complete.
- Obituary data obtained from a third-party vendor.

Approximately half of the death dates in Market Clarity are from the SSA DMF and a third from obituaries; the remaining data sources account for 20% of dates.

Approval through the Optum data disclosure analysis process is required for the use of exact death dates. Exact death dates are not used for this study; death dates based on month and year are used for generalizability to other claims databases.

4.3. Study Sample

4.3.1. Patient Identification

The study sample comprises patients with newly diagnosed multiple myeloma who have both claims and EHR data in Market Clarity. The observation period is from 1 July 2015 to 30 September 2022. The identification period for patients with newly diagnosed multiple myeloma is 1 January 2016 to 31 March 2022. Feasibility results, based on Market Clarity data, are presented in Appendix A. The identification period in the feasibility was shorter

than the identification period specified in this document because more recent data are available since the feasibility was generated.

4.3.1.1. Inclusion Criteria

Patients are eligible for the project sample if they meet the five inclusion criteria listed below.

- **Newly diagnosed MM:** ≥ 2 non-diagnostic medical claims for multiple myeloma (ICD-9-CM=203.00 or ICD-10-CM code=C90.00) in any position on claim, on separate days within 30 days, during identification period. The index date is the date of the first medical claim for multiple myeloma. Diagnosis codes for MM are in Appendix B, tab "MM dx code". A non-diagnostic claim refers to a claim for treatment of a condition (e.g., infusion) rather than diagnosis. For example, claims for diagnosis of a disease (e.g., procedures like biopsies and mammograms), would not be included.
- **Adults:** ≥ 18 years in index year.
- **Baseline continuous enrollment (CE) of six months:** Six months of CE with medical and pharmacy benefits before the index date. The baseline period is fixed and is the six months before the index date. Patients with gaps in enrollment of ≤ 32 days are considered to be continuously enrolled.

The purpose of the baseline in this project is to create a period during which patients who were previously diagnosed with MM are identified; these patients will be excluded at a later stage (see Section 3.3.1.2 Exclusion Criteria).
- **Follow-up CE of at least six months:** ≥ 6 months of CE starting on the index date and ending on the earliest of death, health plan disenrollment, or the study period end for patients who do not have an assigned date of death < 6 months after the index date. For patients with an assigned date of death < 6 months after the index date, the follow-up period ended at the assigned date of death.
- The minimum of 6 months was defined by GSK and Optum because the initial MM LOT is expected to last for ≥ 6 months to achieve complete remission.

- As noted in above, mortality data included month and year of death. The assigned date of death for data set construction was set to the 15th of the month.

- **Evidence of treatment with agents for the purpose of treating MM based on FDA-approval, NCCN guidelines, and peer-reviewed literature during the follow-up period:** Patients will be required to have evidence of an FDA-approved MM oncology therapy in claims during the follow-up period. These will be identified with pharmacy claims (NDC) and medical claims (CPTs and HCPCS) codes.

Receipt of corticosteroids only (e.g., dexamethasone) will not be considered evidence of receiving an approved MM oncology therapy. NDC codes and HCPCS codes for approved MM therapies are in Appendix B, tabs "MM approved therapy – NDC" and "MM approved therapy – HCPCS", respectively.

4.3.1.2. Exclusion Criteria

Patients meeting any of the criteria described below are excluded from the study sample.

- **Evidence of systemic oncology therapy in claims or EHR during the baseline period:** For claims, ≥ 1 medical or pharmacy claim for systemic oncology therapy during the baseline period; for EHR, ≥ 1 oncology therapy (from Prescription Written or Medication Administered EHR tables) or procedure (from Procedure EHR table) record for an oncology therapy. The systemic oncology therapies are listed in Appendix B, tab "Oncology Therapies".
- **Evidence of any MM (each of newly diagnosed and R/R MM), during the baseline period:** ≥ 1 non-diagnostic medical claim for MM (ICD-9-CM=203.00, 203.01, 203.02) or ICD-10-CM code=C90.00, C90.01, C90.02) during the baseline period. Diagnosis codes for MM are in Appendix B, tab "MM dx code".
- **Evidence of another cancer in the baseline period:** Patients meeting any of the following three conditions will be excluded: 1) ≥ 2 non-diagnostic ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type; 2) ≥ 2 non-diagnostic ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the a metastatic cancer; or 3) ≥ 1 ICD-9-CM or ICD-10-CM code, within 30 days,

for a primary cancer and ≥ 1 ICD-9-CM or ICD-10-CM code for a metastatic cancer. Codes (ICD-9-CM and ICD-10-CM) are in Appendix B, tab "Other malignancies".

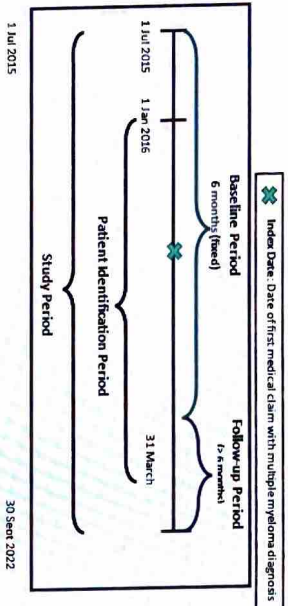
- **Evidence of pregnancy:** ≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the baseline or follow-up period. Codes for pregnancy are in Appendix B, tab "Pregnancy".

- **Clinical trial participation:** Evidence of clinical trial participation during each of the baseline and follow-up periods will be flagged. Codes for clinical trial participation are in Appendix B, tab "Clinical trial".

4.3.2. Observation Period

All patients must have a minimum of 12 months of CE in the health plan: 6 months of baseline and at least 6 months of follow-up (or less, in the event of death). The study periods are presented below in Figure 1.

Figure 1. Study observation periods



4.3.3. Confidentiality

No patient's identity or medical records will be disclosed for this study except in compliance with applicable law.

4.4. Line of Therapy Algorithm

LOTs are not explicitly captured in claims data and therefore an algorithm based on receipt and timing of therapy is implemented. This section describes the algorithm GSK and Optum developed to identify lines of therapy and how the variables in the algorithm are derived. The variables and variable definitions for the LOT algorithm are informed by the clinical literature, input from an Optum expert hematologic oncologist consultant, and the Optum/GSK study team.

The LOT algorithm has been applied to real-world data. As described in Section 2, GSK and Optum have worked through an initial (first) and revised (second) iteration of the MM

LOT algorithm. This start-up document has been an iterative document, now updated to reflect revisions to the LOT algorithm.

For each patient, up to five lines of therapy are identified for the entire follow-up period. The LOT variables are described immediately below and the analytic methods for the LOTs, including the LOT validation, are described in Section 5.2.

The MM therapies are identified from procedure codes (e.g., infusions) in medical settings, including ambulatory care settings and hospitals, and prescriptions found in claims. The medications and associated codes used in the LOT algorithm are in Appendix B, tabs "MM approved therapy – NDC", "MM approved therapy – HCPCS", and "Oncology therapies". All rules described below are based on oncology therapies only (e.g., not corticosteroids).

Note that the same rules and definitions may be described in multiple places. This redundancy is to provide the reader with the information to understand and interpret the specific rule being described without needing to refer to other sections of the document.

4.4.1. LOT Inputs

- **MM therapies:** FDA-approved MM oncology therapy as identified from FDA, NCCN guidelines, and peer-reviewed literature. NDC codes and HCPCS codes for approved MM therapies are in Appendix B, tabs "MM approved therapy – NDC" and "MM approved therapy – HCPCS", respectively.
- **Hematopoietic stem cell transplant (HSCT):** A LOT-specific binary variable is created to classify whether patients have an HSCT during the LOT. HSCT is operationalized as ≥ 1 non-diagnostic procedure code for an HSCT. Patients can have multiple claims for a single HSCT event, which includes claims for the HSCT harvesting and potential adverse events related to the HSCT. For patients with consecutive HSCT claims for up to 14 days, the HSCT date is the date of the last claim for HSCT. For patients with more than 14 days of consecutive HSCT claims, day 14 will be considered the HSCT event because later claims may be for treatment of adverse events associated with HSCTs.
- **Planned tandem HSCT:** Tandem HSCTs will be considered planned if two autologous HSCTs occur ≥ 60 to ≤ 180 days apart; otherwise, they will be considered unplanned.²⁰

Presence of an unplanned HSCT will end the current LOT the day before the unplanned HSCT if the LOT has not otherwise ended prior to the date of the unplanned HSCT.

- **Tandem HSCT:** Patients will be classified as having a tandem HSCT if they receive at least two autologous HSCTs ≥ 60 to ≤ 180 days apart. An autologous HSCT followed by an allogeneic HSCT will not be considered planned because an allogeneic HSCT typically follows failed autologous HSCTs.

- **Allogeneic HSCT:** Presence of an allogeneic HSCT immediately ends a LOT and starts a new LOT. The new LOT will consist entirely of the allogeneic HSCT and no other MM therapies.

- **Conditioning regimen:** A conditioning regimen is a period of any length in days during which only a valid conditioning medication(s) is available. A conditioning regimen may occur at any point during a LOT except after an HSCT in the same LOT. The MM conditioning therapy medications are melphalan monotherapy, cyclophosphamide monotherapy, or melphalan+cyclophosphamide dual therapy.

If a conditioning regimen interrupts the initial regimen (see "Maintenance therapy" below for definition of initial regimen) but is not directly followed by HSCT and the initial regimen resumes, then the LOT does not end. This addresses the real-world data that indicates a patient may start a conditioning period, only to have the conditioning period discontinue without an HSCT and the patient's initial regimen resumes.

Once a conditioning regimen has begun, it continues until discontinuation (see below for discontinuation definition), presence of an HSCT, right censoring, or the addition of any MM therapy automatically causes the conditioning regimen to end.

- **Maintenance regimen:** A maintenance regimen is a period of 120 days^a or longer during which only a valid maintenance therapy is available (see list below of valid mono/dual maintenance therapies). Furthermore, the maintenance therapy must have been a part of the LOT's initial regimen (for LOT1 the initial regimen includes

^a If an otherwise valid maintenance period that came after a conditioning period that did not result in an HSCT or an otherwise valid maintenance period that came after an HSCT did not reach 120 days due to follow-up ending, then that period is still assigned as maintenance.

all MM therapies identified during the first 60 days of LOT1, for LOT2-5 the initial regimen is defined as all MM therapies identified during the first 30 days of the LOT). A maintenance regimen may occur in one of three situations during a LOT:

1. The LOT's initial regimen transitions into a maintenance regimen, such that non-maintenance medications present during the initial regimen are discontinued leaving only maintenance medications. For example, a patient has an initial regimen of cyclophosphamide and daratumumab, and then discontinues cyclophosphamide leaving only daratumumab, which is a valid mono maintenance per the list below.
2. Following a conditioning regimen that did not result in an HSCT.
3. Following an autologous HSCT (with 180 days of the HSCT or in the case of a tandem autologous HSCT within 180 days of the second HSCT). The expectation after an HSCT is for a maintenance period and thus instead of the usual 120 days, we check for a valid 30-day (minimum) maintenance period

- a. If a full 30-day period is unable to be observed due to end of follow-up, we will still count whatever days are available as a maintenance period to avoid a new LOT starting immediately before the end of follow-up.

The following will be considered maintenance therapies, per NCCN Guidelines²⁶.

- Lenalidomide monotherapy (NCCN Preferred Regimen for both transplant and non-transplant candidates)
- Bortezomib monotherapy (NCCN Other Recommended Regimen for both transplant and non-transplant candidates)
- Daratumumab monotherapy (NCCN Other Recommended Regimen for transplant candidates)
- Ixazomib monotherapy (NCCN Other Recommended Regimen for both transplant and non-transplant candidates)
- Bortezomib/lenalidomide dual therapy (NCCN Useful in Certain Circumstances, for non-transplant candidates)
- Carfilzomib/lenalidomide dual therapy (NCCN Useful in Certain Circumstances for transplant candidates)

Analysis of real-world data consistently shows that patients may receive regimens that are inconsistent with NCCN guidelines. For example, although daratumumab is recommended for transplant candidates only, it is possible that patients who do not undergo an HSCt will receive this medication. Therefore, any of the maintenance medications listed above were considered maintenance medications for all patients, regardless of transplant status.

Once a maintenance regimen has begun (e.g., observing a valid maintenance medication(s) by itself for at least 120 days), it continues until discontinuation (defined below), presence of an HSCt, right censoring, or the addition of any MM therapy automatically causes the maintenance regimen to end. In most cases, the end of a maintenance period will also mark the end of a LOT except when the maintenance period is immediately followed by a conditioning period.

4.4.2. LOT Variables and Parameters

- **LOT start date:** LOT1 begins on the date of the first fill for an MM therapy following a patient's index date. Subsequent LOTs will begin on the earlier of an HSCt or the date of the first fill for an MM therapy following the end of the previous LOT. MM therapies are medications used to treat MM that are FDA-approved, based on NCCN guidelines, and/or reported in the peer-review literature.
- **LOT regimen:** A LOT regimen includes all MM therapies received within 60 (30) days¹⁶ on and following the start of LOT1 (LOT2-5) (see LOT start date above). For LOT1, the first therapy received is considered the induction therapy.
- **LOT end date:** The LOT will continue until the earliest of any of the following:
 1. **Addition or substitution** of a new MM treatment from a different drug class after the initial agent(s) (exception: conditioning MM treatments discussed below). Change to another medication within the same class of medication would likely occur because of toxicity and not because of progression. There is an exception to this rule:
 - Change from lenalidomide to pomalidomide, which are medications in the same class, will trigger the end of the line of therapy. This assumes that the change in medication is because of progression (e.g., lenalidomide refractory) but patients may change from lenalidomide to pomalidomide because of toxicity

which cannot be identified in the claims-based analysis. Therefore, changes from lenalidomide to pomalidomide will trigger the end of the LOT.

2. **Discontinuation of all agents in the regimen:** Discontinuation is defined as a treatment gap after the run-out date of all MM treatments in the LOT. Run-out date is the last date that a medication for a specific prescription is considered to be at hand / available.

- A 90-day treatment gap is observed in the LOT algorithm.¹⁷
- The end date is the run-out date before the gap. The run-out date is calculated using a patient's combined infused/injected drugs as well as prescribed drugs. Overlapping pharmacy benefit drug prescriptions "pushed out" the run-out date. See the LOT1 Specifications Excel document tab "DS_2 PROCESSING" for more details.

- The days supply on medical claims is always missing and therefore a 30 days supply was assumed. When pharmacy claims were missing days supply, or when days supply values were anomalous, the claims were excluded.

- If a LOT ends due to discontinuation and an identical regimen starts 90 or more days later without any other intervening regimens, maintenance periods, or HSCts, it will not be considered a new line.²⁰

3. **Unplanned HSCts:** If an autologous HSCt occurs > 180 days following a previous autologous HSCt in the LOT, then the LOT will end the day before the latter HSCt. Any allogeneic HSCts are considered a new LOT and the current LOT will end the day before an allogeneic HSCt occurs. Single autologous HSCts and tandem HSCts are considered a continuation of the line of therapy.

4. **HSCts not followed by maintenance:** If an HSCt is not followed by a maintenance regimen within 180 days, then the last day of the LOT is the date of the HSCt.

5. Death
6. Health plan disenrollment
7. End of the study period

8. End of maintenance regimen: The LOT will end after a maintenance regimen ends due to:

- The introduction of any MM therapies including therapies that were part of the original regimen (e.g., a LOT has an initial regimen of Bortezomib+Lenalidomide and later the patient has a Bortezomib mono maintenance regimen; if Lenalidomide restarts at some point then the maintenance regimen will end as we observed 120+ days of only Bortezomib and the reintroduction of Lenalidomide could indicate progression).
 - i) Exception: Prior to a patient receiving an HSCt in a given LOT, a maintenance regimen may be interrupted by conditioning regimen without causing the LOT to end.
 - Discontinuation: Defined similarly to above; a 90-day treatment gap after the run-out date of the MM therapies(s) in the maintenance period. For patients with planned tandem HSCtS followed by a maintenance regimen, a discontinuation date will be sought only after their second HSCt.
 - i) Exception: Prior to a patient receiving an HSCt in a given LOT, a maintenance regimen that discontinues only to have a conditioning period start within 90 days will not cause the LOT to end.
 - ii) Exception: if an HSCt has not yet occurred during the LOT and an HSCt occurs within 60 days of the maintenance regimen run-out date then the LOT will not end.
 - Unplanned HSCtS (defined above), Death, Health plan disenrollment, end of study period.
- ## 9. End of conditioning regimen: The LOT will end after the conditioning regimen ends due to:
- The introduction of any MM therapies that does not result in a maintenance regimen.
 - Discontinuation: Defined similarly to above; a 90-day treatment gap after the run-out date of all MM conditioning treatments in the conditioning period.

i) Exception: if an HSCt has not yet occurred during the LOT and an HSCt occurs within 60 days of the conditioning regimen run-out date then the LOT will not end.

- Unplanned HSCtS (defined above), Death, Health plan disenrollment, end of study period.

Events that will not end the LOT are:

- **Single autologous HSCt or tandem hematopoietic stem cell transplants (HSCt) ≤ 180 days apart:** HSCtS are usually preceded by induction therapy and followed by consolidation and maintenance therapy. Tandem HSCtS are two HSCtS ≥ 60 to ≤ 180 days apart. These are considered planned and a continuation of the line of therapy. Codes for HSCtS are in Appendix B, tab "HSCt".
HSCtS must be within 60 days after the end of the induction, conditioning, or maintenance regimen to be considered a continuation of the LOT. This 60-day rule was recommended by Optum's KOL^b and informed by Charalampous et al.'s findings that the median time from end of induction to transplant was 33 days (interquartile range=27-42).¹⁸

- **The start of a conditioning regimen:** Receipt of a valid conditioning regimen (see 4.4.1. for definition) following or interrupting a patient's initial regimen will not end the LOT.
- **The start of a maintenance regimen:** Receipt of a valid maintenance regimen (see 4.4.1. for definition)

4.4.3. LOT Classifications

- **First-line (L1; LOT1):** The first line of therapy received by patients with newly diagnosed MM. This includes induction therapy and may also include maintenance and HSCtS.

^b Optum's KOL is Dr Allan Stawis.

Second-line (L2) and later LOTs: Indicated by initiation of new therapy or an HSCT after the end of the previous LOT (1L +/- maintenance or 1L HSCT + maintenance). The LOT rules described above apply to LOT2 and later LOTs with the caveat that for LOT2 and later LOTs the regimen identification window (to determine which MM treatments are present in the LOT) is 30 days instead of 60 days

- CAR T: No patients in the final sample were found to be receiving CAR T treatment.

5 Bias

5.1. Limitations

- LOTs are not explicitly captured in claims data and the algorithms used to identify lines of therapy are based on multiple assumptions.
- Patients with evidence of MM during the baseline will be excluded from the study sample to ensure that the patients ascertained during the identification period are newly diagnosed with MM. However, because ICD-9-CM and ICD-10-CM codes do not exist to differentiate smoldering from active MM, this will result in the exclusion of patients with smoldering MM who progress to active MM. Therefore, the LOT algorithm developed may not be generalizable to patients who have smoldering MM who progress to active MM.
- The medical records that contribute to the electronic health record (EHR) database may not represent the entire patients' care history because they may have received care at clinics or hospitals outside those in the EHR database. The presence of encounters with a health care provider and flags for EHR records at the beginning and end of claims-based LOTs will be used as a potential marker of the completeness of the records.
- Medications with blocked information (e.g., during the time that health insurers are identifying payment and authorization processes) may be missing from Market Clarity

- This study's results will be limited by the detail and quality of the structured fields in the databases.

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7 Appendices

7.1. Appendix A. Original Feasibility

Please note that these counts were generated during the proposal process and are for information purpose only.

Criteria	Number of patients
Inclusion	
Patients with at least 1 non-diagnostic claim for multiple myeloma (ICD-9-CM=203.00 or ICD-10-CM=C90.00) during 1 January 2016 to 31 December 2021 (the identification period). The first date is the index date	95,796
Continuous enrollment: 6 months before index date (Baseline period)	35,938
Continuous enrollment: ≥3 months after index date, starting on index date (Follow-up period)	32,945
Age ≥18 years as of the index date	32,825
At least 1 claim for systemic cancer therapy on or after the index date. The date of the first claim is the first time of therapy start date	18,174
No evidence of systemic cancer therapy in baseline period	11,958
No evidence of MM in baseline period	7,971
No evidence of R/R MM (≥2 claims on different dates of service), in baseline period	7,864
No evidence of other cancers in baseline period	5,911

Counts among 5,911 patient population	
Patients with information on prescriptions written/administered in EHR	
Patients with at least 1 non-missing prescriptions written or medications administered occurring on or after LOT1 start date	2,467
Patients with at least 5 non-missing prescriptions written or medications administered on different dates of service occurring on or after LOT1 start date	1,853
Patients with physician description of treatment rationale in EHR	
Patients with at least 1 non-missing discontinuation reason (e.g., "lack of efficacy", "side effect", "alternate therapy prescribed") occurring on or after LOT1 start date	1,685
Patients with at least 2 non-missing discontinuation reasons (e.g., "lack of efficacy", "side effect", "alternate therapy prescribed") on different dates of service occurring on or after LOT1 start date	1,506
Patients with physician description of drug action in EHR	
Patients with at least 1 non-missing drug action (e.g., "start", "stop", "switch") occurring on or after LOT1 start date	2,003
Patients with at least 2 non-missing drug actions (e.g., "start", "stop", "switch") on different dates of service occurring on or after LOT1 start date	1,730

7.2. Appendix B. Code Lists

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7.3. Appendix C. Pressure-testing Patient Identification Criteria

Additional steps, summarized below, were implemented to confirm patients' MM diagnosis and to identify and exclude those with relapsed MM. These steps were supplemental to the inclusion and exclusion criteria and, ultimately, did not change the final sample size of N = 5,372 patients.

- **EHR flags for MM diagnosis date:** Patients meeting the following three criteria in the EHR diagnosis table were be flagged: 1) diagnosis of ICD-9-CM=203.00 or ICD-10-CM code=C90.00; 2) diagnosis date +/- 30 days of the index date; 3) diagnosis status of "possible diagnosis of", "diagnosis of", "not recorded".
10. This step was used to confirm the claims-based diagnosis of newly diagnosed MM.
- **Claims flags for longer baseline periods:** Patients with each of 9, 12, and 18 months of baseline continuous enrollment were flagged. These longer baseline periods were considered to maximize identification of patients with MM patients before the identification period and determine the resulting sample size after these patients were excluded. GSK and Optum decided on a 6-month baseline period.
- **Claims flag for minimum 3-month follow-up period:** GSK and Optum explored a minimum 3-month follow-up period for all patients and decided on the minimum 6-month variable follow-up except for patients who died less than 6 months after the index date.
- **EHR flags for approved MM therapies:** Patients with ≥ 1 EHR record for approved MM therapies from the EHR prescriptions written, medications administered, and procedure tables were flagged. Approved MM therapies in the EHR tables were measured with the same NDC, CPT, and HCPCS codes applied to the claims data. These flags were used to identify the potential number of patients available for the EHR validation of the claims-based LOT.

- **Flags for baseline MM diagnoses:** Patients with ≥ 1 non-diagnostic medical claim for MM during 9-month, 12-month, and 18-month baseline periods, patients with ≥ 1 EHR diagnosis table record with a diagnosis of MM during 6-month, 9-month, 12-month, and 18-month baseline periods, and patients with ≥ 1 EHR diagnosis table record with diagnosis_status = "possible diagnosis" of MM during 6-month, 9-month, 12-month, and 18-month baseline periods were identified.
- 11. Exclusion of patients with baseline MM may have excluded those with smoldering MM, which can be a precursor to active MM: a specific diagnosis code does not exist to differentiate smoldering from active MM. The implication of this exclusion is described in the limitations section.
- **Flags for baseline R/R MM:** Patients with ≥ 1 non-diagnostic medical claim for R/R MM during 9-month, 12-month, and 18-month baseline periods, patients with ≥ 1 EHR diagnosis table record with a diagnosis of R/R MM during 6-month, 9-month, 12-month, and 18-month baseline periods, and patients with ≥ 1 EHR diagnosis table record with diagnosis_status = "history of" of R/R MM during 6-month, 9-month, 12-month, and 18-month baseline periods were identified.
- **EHR flags for other baseline cancers:** Patients with ≥ 2 EHR diagnosis table records on separate dates and within 30 days of each other for the same primary tumor type, for metastatic cancer, and for a primary cancer and metastatic cancer were identified.
- **Flags for claims and EHR records with diagnoses for other cancers during follow-up:** Patients with ≥ 2 medical claims or EHR diagnosis table records on separate dates and within 30 days of each other for the same primary tumor type, for metastatic cancer, and for a primary cancer and metastatic cancer were identified. These patients were identified because the LOT algorithm was developed for patients with MM only; those with other cancers may have received different treatments. GSK and Optum decided not to exclude patients with other cancers in the follow-up period.

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